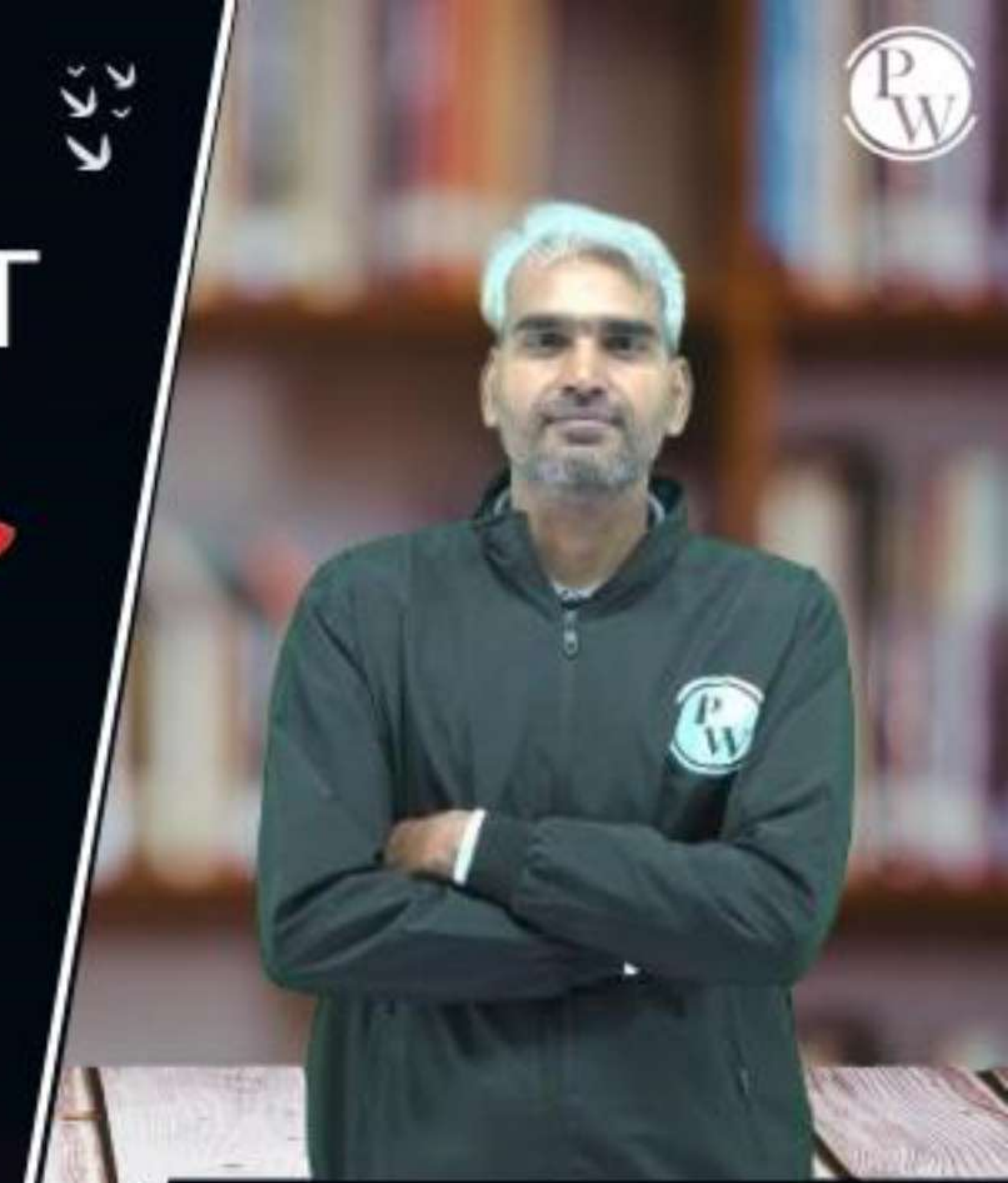
Computer Science & IT

Discrete Mathematics

Set Theory & Algebra

Lecture No. 06



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Recap of Previous Lecture

- Topic — Irreflexive relation 2^{n^2-n}
- Topic — Symmetric relation $2^n * 2^{\frac{n^2-n}{2}} = 2^{\frac{n^2+n}{2}}$
- Topic — Anti-symmetric relation $2^n \cdot 3^{{}^nC_2} = 2^n \cdot 3^{\frac{n^2-n}{2}}$
- Topic — Asymmetric relation $3^{\frac{n^2-n}{2}}$

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Topics to be Covered

Topic Transitive relation

Topic Complement of a relation

Topic Inverse of a relation

Topic Composite of two relations

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Note :→ ① Any subset of an anti-symmetric relation is also an anti-symmetric relation

② Any subset of an asymmetric relation is also an asymmetric relation

③ Every asymmetric relation is an anti-symmetric relation, but Every anti-symmetric relation need not be asymmetric

<u>Note</u>:- ① Empty relation {i.e. { }} is symmetric, anti-symmetric as well as asymmetric

② Diagonal relation on any set A is the largest relation which is symmetric as well as anti-symmetric

Topic : Transitive Relation

A relation R on set A is called a transitive relation if and only if,

$\left(\text{if } a R b \text{ and } b R c, \text{ then } a R c\right), \forall a, b, c \in A$

i.e, $\left(\text{if } (a, b) \in R \text{ and } (b, c) \in R, \text{ then } (a, c) \in R\right), \forall a, b, c \in A$

ends with 'b'

starts with 'b'

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eg:- Let $A = \{1, 2, 3\}$

① $R_1 = \{\ \}$ is the smallest transitive relation on set A

② $R_2 = A \times A$ is the largest transitive relation on set A.

③ $R_3 = \{(1,1), (2,2)\}$ is a transitive relation

$$\underbrace{(1,1) \ \& \ (1,1) \ \therefore (1,1)}_{\text{all cond}^n \text{ are true}}$$

④ $R_4 = \{ (1,2), (2,1), (1,1) \}$

- $(1,2)$

ends with '2' ∴ look for order pairs that starts with '2' $(2,1)$

$(1,2)$ & $(2,1)$ ∴ $(1,1)$ must be present in relation

We know $(1,1)$ is present ∴ No problem

- $(2,1)$

ends with '1' ∴ look for order pairs that starts with '1' $(1,2)$ & $(1,1)$

$(2,1)$ & $(1,2)$ ∴ $(2,2)$ must be present in the relation

We know $(2,2) \notin R_4$ ∴ R_4 is not transitive

$(2,1)$ & $(1,1)$ ∴ $(2,1)$ must be must be present

and it is already present

Note: (a, b) & (b, b) $\therefore$ (a, b) must be present

Comparison of diagonal order pair with any other order pair will never result in a new order pair w.r.t. transitivity property

i.e. presence of diagonal order pair will never create any problem w.r.t transitivity

But some times absence of some diagonal order pair may create problem w.r.t. transitivity.

eg. $A = \{1, 2, 3\}$

$R_5 = \{(1,2), (1,3)\}$

Ends with '2',
but no
order pair
starts with '2'
∴ if Cond^n
it self is
false
Hence,
No problem

Ends with '3',
but no order pair
starts with '3'.
∴ If Cond^n itself
is false.
Hence, No problem

Hence R_5
is transitive

eg: $A = \{1, 2, 3\}$

$R_6 = \{ (2,3), (1,2) \}$

Ends with '3'
No order pair starts with '3'

Ends with '2'
and (2,3) starts with '2'

$(1,2)$ & $(2,3)$ $\therefore$ $(1,3)$ must be present

but $(1,3) \notin R_6$

$\therefore$ R_6 is not transitive

Topic : Complement of a relation

Complement is always defined w.r.t. Universal set.

$A^c = U - A$

* Let 'R' is a relation from set A to set B, the compement of relation R is denoted by R^c and it is defined as.

$$R^c = \underbrace{A \times B}_{\text{Universal relation}} - R$$

eg: $A = \{1, 2\}$ & $B = \{a, b, c\}$
R is a relation from A to B defined as $R = \{(1,b), (1,c), (2,b)\}$

then, $R^c = A \times B - R$

$= \{(1,a),(1,b),(1,c),(2,a),(2,b),(2,c)\} - \{(1,b),(1,c),(2,b)\}$

$= \{(1,a), (2,a), (2,c)\}$

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Topic : Inverse of a relation

- Let R is a relation from set A to set B.
Inverse relation of relation R can be denoted by R^{-1}.
and R^{-1} will be a relation from set B to set A
which is defined by

$$R^{-1} = \{ (b, a) \mid (a, b) \in R \}$$

from set B

from set A

$\therefore$ from set B to set A

R is from A to B
$\therefore$ in order pair (a, b)
a is from set A
& b is from set B

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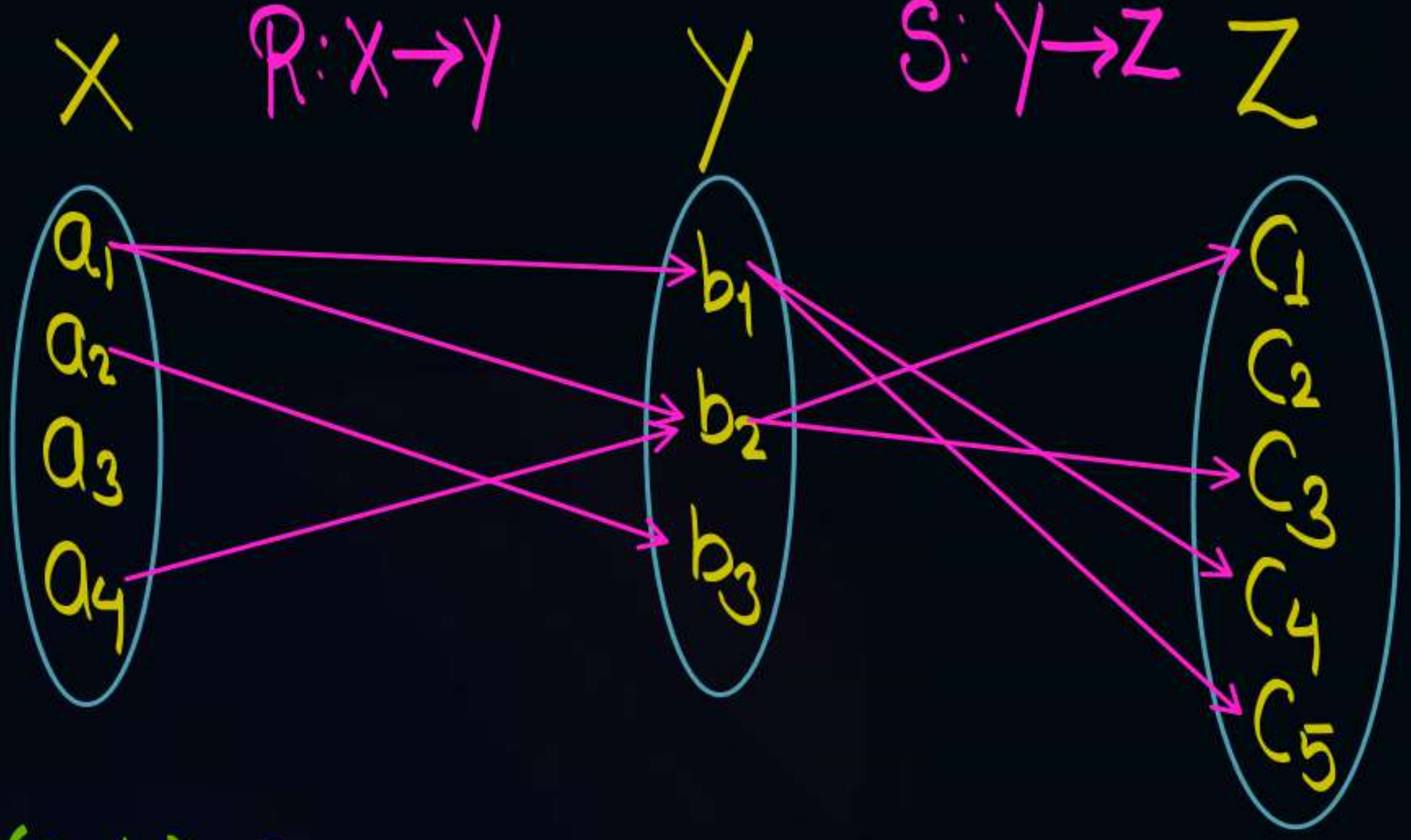
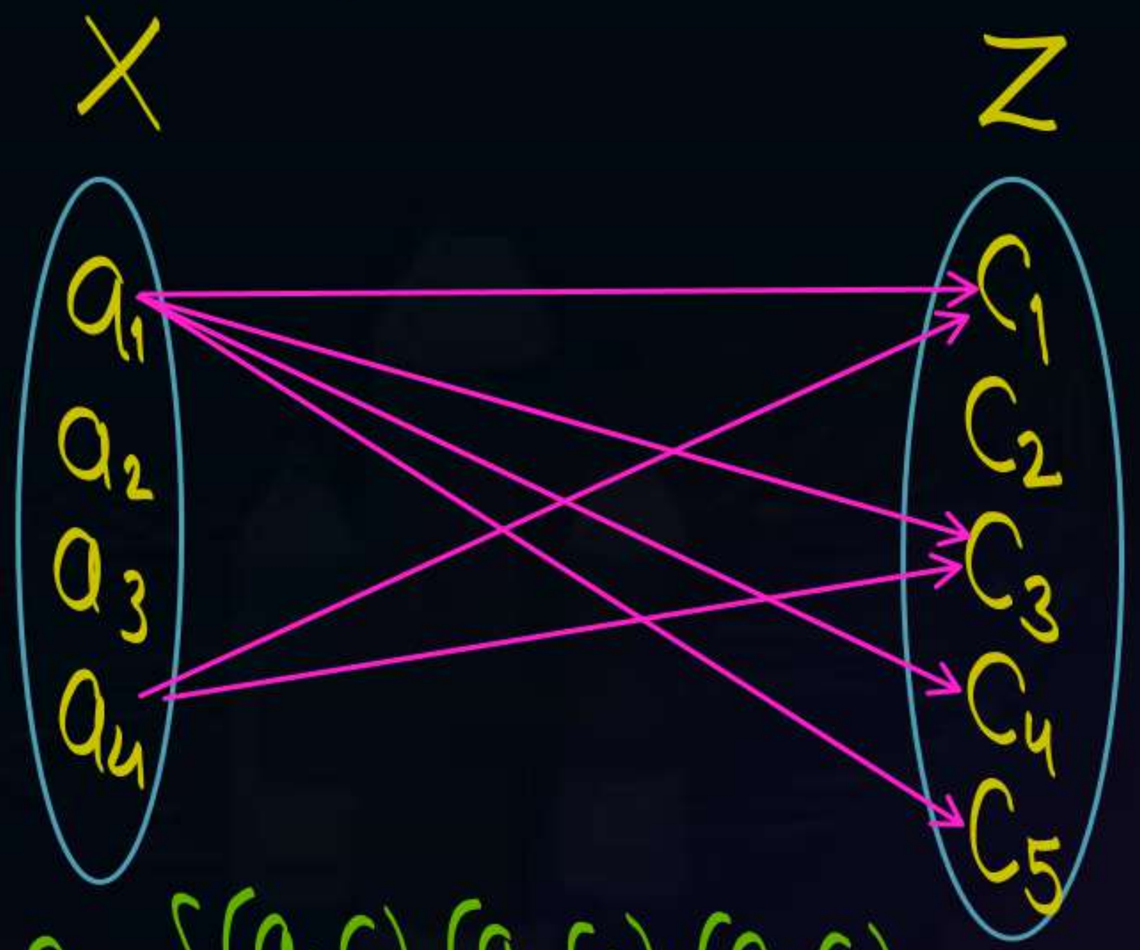
Topic : Composite of two relations

Composite of relation R & S i.e. R;S is a relation from X to Z

$R;S : X \longrightarrow Z$





$R = \{(a_1, b_1), (a_1, b_2), (a_2, b_3), (a_4, b_2)\}$

$S = \{(b_1, c_4), (b_1, c_5), (b_2, c_1), (b_2, c_3)\}$

$R;S = \{(a_1, c_1), (a_1, c_3), (a_1, c_4), (a_1, c_5), (a_4, c_1), (a_4, c_3)\}$

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Topic : Composite of two Relations

Let X,Y and Z be three sets, and

If R is a relation from X to Y, and S is a relation from Y to Z, then their composition R;S is the relation defined as,

or simply "RS"

R;S = {(x,z) | (x,z) ∈ X × Z and there exists y ∈ Y such that (x,y) ∈ R and (y,z) ∈ S}

it states that R;S is a relation from X to Z

i.e., R;S is a relation from X to Z defined by the rule that (x,z) ∈ R;S if and only if there is an element y ∈ Y such that (x,y) ∈ R and (y,z) ∈ S

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Note: Composite of relation R and S need not be Commutative

i.e $R;S \neq S;R$

* It may be case that when $R;S$ is defined then $S;R$ is not defined

2 mins Summary

Topic Transitive relation ✓

Topic Complement of a relation ✓

Topic Inverse of a relation ✓

Topic Composite of two relations ✓

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THANK - YOU

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